PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA Div: B **Semester: III**	**Course Code: MCA01604** **Course Name: Data Science Laboratory**	**Batch: S2**
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CO No: CO605.3	**Assignment No: 10**	

Title: Create Bar Chart-Stacked taking X axis as Month and Y axis as Revenue and add Legend

- colors <- c("green", "orange", "brown")
- months <- c("Mar", "Apr", "May", "Jun", "Jul")
- regions <- c("East", "West", "North")

Code:

```
import matplotlib.pyplot as plt
import numpy as np

colors = ["green", "orange", "brown"]
months = ["Mar", "Apr", "May", "Jun", "Jul"]
regions = ["East", "West", "North"]

revenue_data = np.array([
  [20, 35, 30, 35, 27],
  [25, 28, 29, 31, 33],
  [40, 23, 38, 33, 45]
])

num_months = len(months)
num_regions = len(regions)

bottom = np.zeros(num_months)

fig, ax = plt.subplots(figsize=(10, 6))

for i in range(num_regions):
    ax.bar(
        months,
        revenue_data[i],
        bottom=bottom,
        label=regions[i],
        color=colors[i],
        edgecolor="white"
    )
```

```
    bottom += revenue_data[i]

ax.set_xlabel("Month", fontsize=12)
ax.set_ylabel("Revenue (in thousands)", fontsize=12)
ax.set_title("Monthly Revenue by Region", fontsize=16)
ax.legend(title="Regions") # Add the legend with a title

plt.tight_layout()
plt.show()
```

Output: